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Hydrogenation

Silylchromium-Catalyzed Selective Semihydrogenation of Undirected Allenes by Sila-Metalation and β-Silyl Elimination

Zheng Luo[+], Linhong Long[+], Shuaiyong Zhao, Xiaoyu Zhang, Yong Peng, Meiming Luo, Hui Chen, and Xiaoming Zeng**

Abstract: The semihydrogenation of simple allenes remains an issue because of the difficulty in controlling of selectivity without coordination assistance from directing groups, wherein over-hydrogenated alkanes and *E*/*Z*-mixed stereo- and regio-isomers can be potentially formed. We report here an electron-rich cyclic (alkyl)(amino)carbene (CAAC)-phosphine-ligated chromium(III) complex serving as precatalyst in promoting the selective semihydrogenation of undirected allenes by forming active silylchromium species. The semihydrogenation allows the selective addition of H_2 to the less substituted and electron-rich alkyl-substituted C=C bonds of allenes while enabling the suppression of competitive over-hydrogenation, proving generally to form trisubstituted *E*-alkenes in high chemo-, regio-, and stereoselectivity. By this CAAC-phosphine-Cr-catalyzed selective semihydrogenation, unsaturated nitro, nitrile, alkenyl, and hydrogenation-sensitive bromide, chloride, and fluoride were compatible with various substitution environments. The in situ forming silylchromium as active catalyst by reaction of low-valent chromium complex with chlorosilane was established by studying the catalytic and stoichiometric reactivity of this species in the semihydrogenation. Deuterium labeling experiments, HRMS, XPS, EPR, and magnetic susceptibility analysis of the related intermediates and theoretical studies support a pathway involving sila-metalation of allene and β-silyl elimination processes with silyl as noninnocent ligand, leading to the *E*-alkene products.

Introduction

Allenes containing 1,2-diene moieties are an important family of unsaturated hydrocarbons in organic synthesis.[1–15] The 3D 1,2-diene structures of allenes are recognized as the bioisosteres of spirocyclobutanes, propargylic scaffolds, and furans with unsaturated cumulative scaffolds. As such, if reducing one of cumulated C=C bonds is appropriately harnessed to encompass a semihydrogenation reaction with H_2, the approach allows the atom-economic formation of substituted alkenes that are particularly important skeletons with broad applications in material science, natural products and pharmaceuticals.[16,17] In principle, the hydrogenation of 1,3-disubstituted allenes could lead to the formation of five compounds, including two regiodifferentiated and two stereodifferentiated semi-hydrogenated alkenes and one fully hydrogenated alkane (Scheme 1a).[18–27] As a consequence, control of the chemo-, regio-, and stereoselectivity of this semihydrogenation remains a challenge, although it would allow to rapidly form valuable 1,1-/1,2-disubstituted and trisubstituted alkene motifs from simple 1,2-diene precursors.[28–35]

To address the selectivity issue, studies can date to the work of Ma and Elsevier, who used a phosphonate and related groups to assist the semihydrogenation of allenic groups by selectively reducing the C_2=C_3 bonds (Scheme 1b).[36] Zhang and Lv extended this semihydrogenation to reduction of allenyl sulfones to allylic motifs by chiral rhodium catalysis.[37] An impressive example of using $Ni(OAc)_2$/Ph-BPE catalyst in achieving selective semihydrogenation of allenyl sulfones was described by Dong and Zhang.[38] These achievements showcase that the directed semihydrogenation of allenic groups with phosphonates and sulfones is a valuable synthetic means for obtaining high selectivity.[39,40] It appeared challenging to reduce one of cumulated C=C bonds of 1,2-diene structures in a selective manner without chelation assistance.

Substituted allenes without a chelating group at two orthogonal cumulative 1,2-diene systems carry the potential in augmenting the space of alkene products that can be accessible, and the method could become increasingly universal for broad impact. However, in the absence of chelation assistance, the regiospecific recognition of one cumulated C=C bond followed by stereoselective hydrogenation without over-hydrogenation is challenging. The study

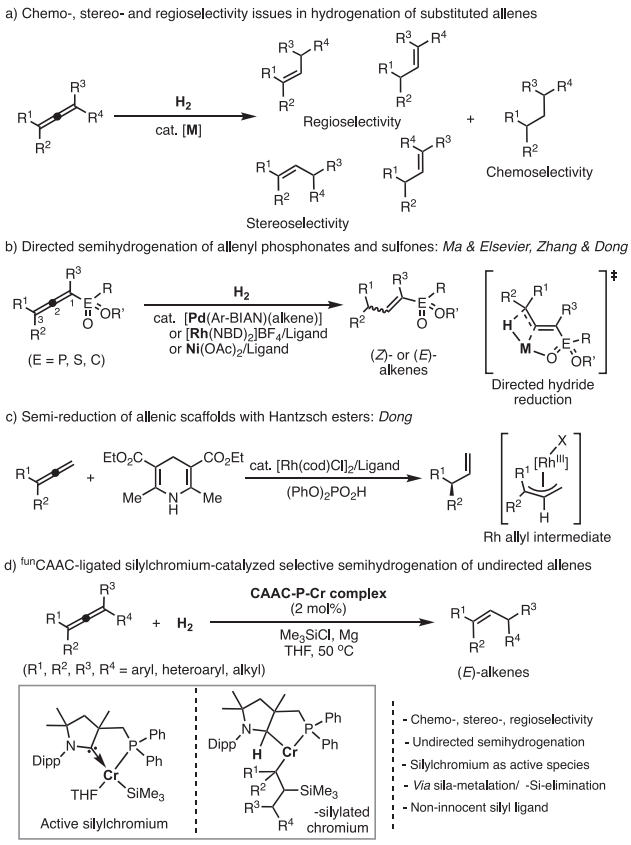
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Scheme 1. Semihydrogenation of 1,2-diene moieties.

by Dong and coworkers showed that if Hantzsch esters were used as H-sources with rhodium catalysis, the selective transfer semihydrogenation of undirected allenes could be achieved in forming terminal alkenes (Scheme 1c).[41] Earth-abundant transition metals have received increasing attention of synthetic community, and their use as catalysts provides the opportunity for disclosing mechanistically distinct reactivity and selectivity in organic transformations.[42] In this context, chromium metal is one of attractive 3*d* metal catalysts for hydrogenation because of its low cost and high natural abundance.[43–48] Shibasaki and Carreira have demonstrated spectacular results by using (aryl)$Cr(CO)_3$ as the catalyst in developing 1,4-semihydrogenation of 1,3-dienes through metallacyclopentene intermediates.[49–51] To achieve the semihydrogenation of 1,2-dienes by reducing one of cumulative C=C bonds, a conceptually and mechanistically new route would require the development of robust metal catalysts and effective catalytic systems. Herein, we describe the chemo-, regio-, and stereoselective semihydrogenation of undirected allenes that was enabled by an electron-rich cyclic (alkyl)(amino)carbene (CAAC)-phosphine-ligated chromium complex with chlorosilane (Scheme 1d). This reaction favors selectively semihydrogenating one of less crowded C=C bonds of 1,2-dienes with 2 mol% catalyst loading, through a CAAC-phosphine ligand and chlorosilane-enabled semihydrogenation by forming silylchromium as active species via sila-metalation and β-silyl elimination to generate *E*-alkenes.

Results and Discussion

Initial Considerations and the Development of Chromium-Catalyzed Semihydrogenation of Undirected Allenes

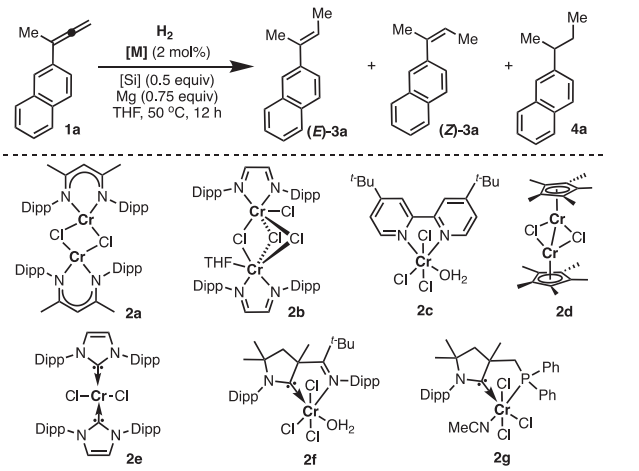
As ligands play important role in enhancement of the reactivity of catalysts by turning the electronic property of metals and altering the steric circumstance around metal centers, a regio-differentiation of cumulated C=C bonds of allenes would depend on the steric hindrance in metal catalysts by preferential coordination with the less crowded ones. After semihydrogenation, the resulting alkenes could be potentially hydrometallated by hydride-metal species to give (alkyl)metal intermediates, which eliminate the β-hydride to regenerate alkenes or are further reduced by hydride in giving over-hydrogenated alkanes.[52–55] To obtain alkenes as products rather than alkanes, the rate for the β-elimination process should be higher than the rate for the hydride reduction. To our knowledge, when a *syn*-coplanar arrangement of the metal–carbon bond and the carbon–silicon bond at the β-position is adopted, the elimination of β-silyl substituent could occur and has been applied in the ruthenium-catalyzed disproportionation by transfer of silyl group to form alkenes.[56] Brookhart's group observed that the elimination of β-silyl group of β-silyl(alkyl) palladium complexes is a facile reaction like the β-hydride and β-methyl abstraction.[57] Daugulis and Brookhart demonstrated that the β-silyl elimination could play an important role in giving key (alkenyl)Pd intermediates for chain transfer in copolymerization reactions.[58] The β-silyl elimination process in nickel-catalyzed silylations of arenes with vinylsilanes was demonstrated.[59] Based on these studies, we wondered whether an analogous β-silyl elimination could be designed and exploited in the selective semihydrogenation of 1,2-diene moieties of allenes to alkenes. In principle, the semihydrogenation can be initiated by a silyl-ligated metal species of [M]–SiR_3 that might be formed in situ by reaction of active metal complex with chlorosilane, and regioselectively adds to the less crowded C=C bond of allene by sila-metalation in forming β-silyl(alkyl) metal intermediate. A hydride reduction and further hydrometalation would allow for access to β-silylated species. We saw the opportunity for forming trisubstituted alkene products in a regio- and stereoselective manner by process of β-silyl elimination in competition with that of β-hydride elimination.[60]

We commenced our studies by exploring the semihydrogenation of disubstituted allene **1a** by using 0.5 equiv of chlorotrimethylsilane as an additive. Under atmosphere of hydrogen, 3*d* transition metal salts such as $FeCl_3$, $CoCl_2$, $NiCl_2$, and $MnCl_2$ as precatalysts do not show any reactivity in promoting the hydrogenation of **1a** with magnesium (Table 1, entries 1–4). We were pleased that commercially available and low-cost 3*d* metal salts of $CrCl_3$ and $CrCl_2$ as precatalysts allowed the reaction to occur sluggishly, albeit giving *E*-/*Z*-mixed stereomers **3a** in around 5% yield and 1:1 stereoselectivity (entries 5 and 6).[61] We examined the effect of ligands on semihydrogenation using their supported chromium complexes as precatalysts. The dinitrogen ligand

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Table 1: Reaction discovery and optimizing conditions.[a)]



Entry	[M]	[Si]	Yield (**3a**)	*E*/*Z*	Yield (**4a**)
1	$FeCl_3$	Me_3SiCl	0%	–	0%
2	$CoCl_2$	Me_3SiCl	0%	–	0%
3	$NiCl_2$	Me_3SiCl	0%	–	0%
4	$MnCl_2$	Me_3SiCl	0%	–	0%
5	$CrCl_3$	Me_3SiCl	4%	50:50	0%
6	$CrCl_2$	Me_3SiCl	6%	51:49	0%
7	**2a**	Me_3SiCl	31%	50:50	0%
8	**2b**	Me_3SiCl	15%	58:42	0%
9	**2c**	Me_3SiCl	8%	55:45	0%
10	**2d**	Me_3SiCl	35%	50:50	0%
11	**2e**	Me_3SiCl	17%	48:52	0%
12	**2f**	Me_3SiCl	67%	99:1	27%
13	**2g**	Me_3SiCl	91% (89%)[b)]	99:1	0%
14	**2g**	Me_2SiHCl	81%	99:1	0%
15	**2g**	Et_3SiH	0%	–	0%
16[c)]	**2g**	$Me_3SiMgBr$	26%	99:1	0%

[a)] Conditions: **1a** (0.4 mmol), **2g** (2 mol%), Mg (0.75 equiv), Me_3SiCl (0.5 equiv), THF (2 mL), H_2 (30 atm), 50 °C, 12 h. ^1H NMR yield was given using dibromomethane as internal standard. The *E*/*Z* selectivity was determined by GC analysis. [b)] Isolated yields in the parenthesis. [c)] Without magnesium.

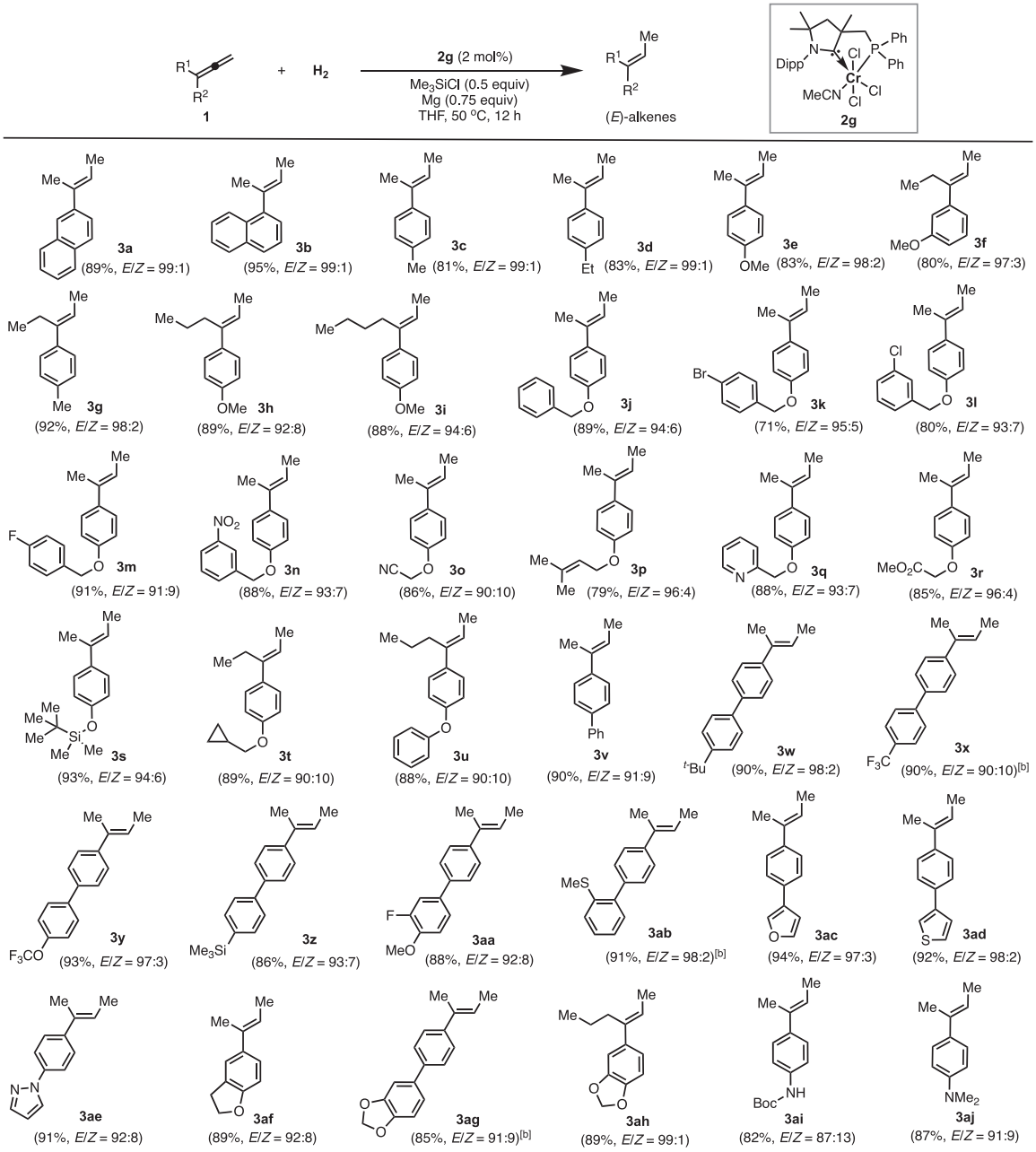
containing β-diketiminato (nacnac) or two imino scaffolds (**2a** or **2b**) enhances the catalytic activity of chromium metal in the semihydrogenation, but gives low *E*/*Z*-selectivity (entries 7 and 8).[62,63] The stereoselectivities are always lower when the use of chromium complexes **2c**–**2e** by ligation with bipyridine, cyclopentene, and normal cyclic (diamino)carbene (NHC), forming **3a** in low conversions (entries 9–11).[64–67] Compared with common NHCs, CAACs show strong nucleophilicity and electrophilicity,[68–70] and their ligated metal complexes have been prepared as robust catalysts in promoting organic reactions.[71–73] The functionalized CAACs (^{fun}CAACs) that contain an imine or phosphine anchor have been used as ancillary ligands for transition metal catalysis by Bertrand and us.[74–77] Gratifyingly, electron-rich and bulky CAAC-imine-supported bidenate chromium complex **2f** displays high reactivity in the semihydrogenation, allowing for the formation of *E*-alkene product (**(*E*)-3a**) in 67% yield and 99:1 stereoselectivity (entry 12).[75,77] However, large amount of over-hydrogenated alkane **4a** was obtained in the reaction (27% yield), resulting in a low chemoselectivity (2.5:1). Interestingly, replacing the imine scaffold by more electron-rich phosphine group in the ^{fun}CAAC ligand (**2g**) fully inhibits the competitive over-hydrogenation, leading to the formation of **(*E*)-3a** in high yield and excellent stereo- and chemoselectivity (entry 13).[76] In addition to Me_3SiCl, the semihydrogenation of **1a** with chlorodimethylsilane proceeds smoothly (entry 14). Whereas the semihydrogenation was fully inhibited using triethylsilane, indicating an important role of chlorosilane in giving active catalyst for the semihydrogenation (entry 15). Despite that chlorotrimethylsilane does not react with magnesium in giving the related Grignard reagent,[78] we prepared the silyl Grignard reagent of $Me_3SiMgBr$ and explored its effect on the semihydrogenation of **1a**. We found that the semi-reduction occurs sluggishly, leading to low conversion in recovery of large amount of allene precursor (entry 16). In the absence of hydrogen atmosphere, we proved that the semi-reduction of allene does not occur with ^{fun}CAAC-Cr catalysis in recovery of 97% of **1a**.

The Scope of CAAC-Phosphine-Chromium-Catalyzed Selective Semihydrogenation of Allenes

Using 2 mol% of CAAC-phosphine chromium complex **2g** combined with 0.5 equiv of Me_3SiCl and 0.75 equiv of magnesium, the selectivity of semihydrogenation was examined in a broad range of allenes with various substitution patterns (Scheme 2). Efficient semihydrogenation was observed over the course of 12 h at 50 °C under atmosphere of hydrogen. The regiochemically pure trisubstituted *E*-alkenes were formed in high yields and chemo- and stereoselectivity with products containing naphthyl (**3b**), methylphenyl (**3c**), ethylphenyl (**3d**), and methoxylphenyl (**3e**) substituents. With ethyl-, propyl-, and butyl-substituted allenes, the semihydrogenation occurs effectively to afford *E*-alkene compounds **3f**–**3i** in 80%–92% yields and 92:8–98:2 stereoselectivity. Because small amount of magnesium was used, halogen substituents of bromide, chloride, and fluoride were tolerated without dehalogenation, emphasizing the mild conditions in achieving high chemoselectivity (**3k**–**3m**). It was noted that easily reducible unsaturated functional groups of nitro (**3n**), cyano (**3o**), alkenyl (**3p**), and pyridinyl (**3q**) by hydrogenation can be retained in the chromium-catalyzed semihydrogenation system. In addition to fluoro substituents, *E*-alkenes that contain medicinally interesting trifluoromethyl and trifluoromethoxy scaffolds can be facilely prepared by the semihydrogenation of 1,1′-disubstituted allenes (**3x** and **y**). Importantly, sensitive methylthio group that usually causes the poisoning of transition metal catalysts in hydrogenation does not decrease the efficiency and selectivity of the semihydrogenation of allene (**3ab**). By using allenes that are substituted by heteroaryl groups such as furanyl (**3ac**), thiophenyl (**3ad**), and pyrazolyl (**3ae**), semihydrogenating the terminal cumulative C=C bonds smoothly occurs without loss of efficiency, leading to the formation of heteroaryl-bearing trisubstituted *E*-alkenes. Allenes containing oxygen-heterocycle groups of 2,3-dihydrobenzofuran (**3af**) and benzo[*d*][1,3]dioxole (**3ag** and **3ah**), and Boc-protected aniline (**3ai**) and dimethylaniline (**3aj**) were hydrogenated effectively by selectively

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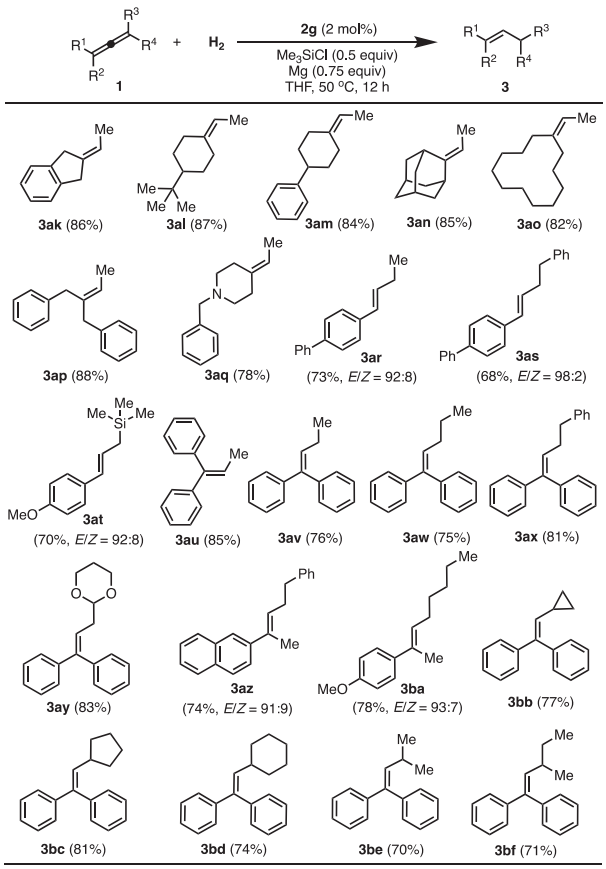
Scheme 2. Selective semihydrogenation of undirected allenes for the synthesis of trisubstituted *E*-alkenes. [a] Conditions: **1** (0.4 mmol), **2g** (2 mol%), Mg (0.75 equiv), Me_3SiCl (0.5 equiv), THF (2 mL), H_2 (30 atm), 50 °C, 12 h. [b] **2g** (3 mol%), 60 °C. Isolated yield. The *E*/*Z* selectivity was determined by GC analysis.

semi-reducing the terminal cumulated C=C bonds, providing access to oxygen-bearing *trans*-selective alkene products.

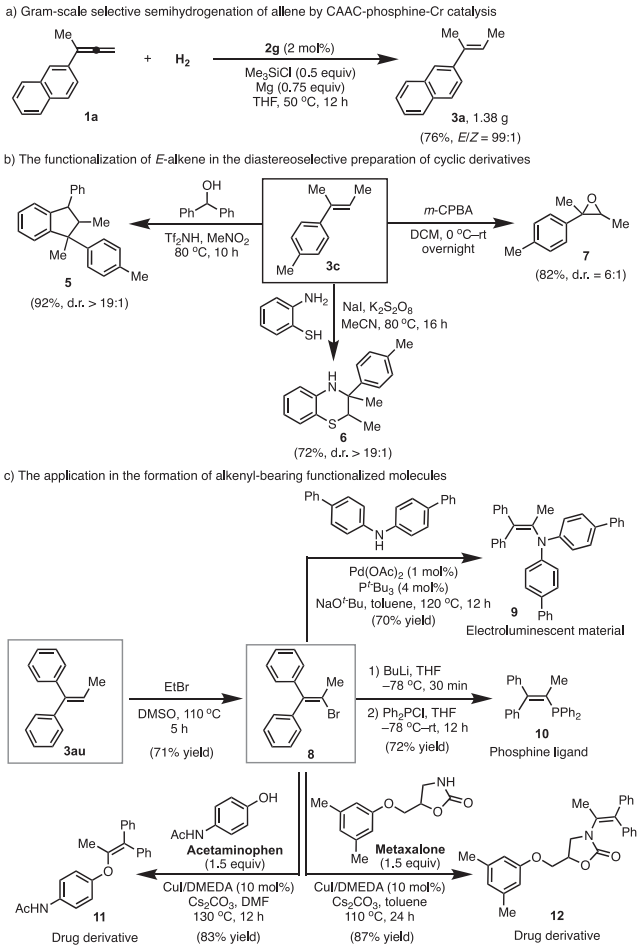
To further investigate the influence of substituents of allenes on the selectivity of semihydrogenation, a range of 1,1′-dialkylated allenes, such as 2-vinylidene-2,3-dihydro-1*H*-indene, 4-vinylidenecyclohexane, 2-vinylideneadamantane, and vinylidenecyclododecane are chosen for catalytic hydrogenation with **2g** (Scheme 3). Importantly, the semihydrogenation forms methyl-substituted alkene products **3ak**–**3ao** without the isomerization of olefins to thermodynamically stable cycloalkenyl compounds. Similarly, the isomerization of the formed alkene does not occur when dibenzyl allene was used in the reduction, providing a valuable method in the preparation of (2-ethylidenepropane-1,3-diyl)dibenzene (**3ap**). Using the semihydrogenation, ethylidenepiperidine derivative **3aq** can be facilely accessible. In addition, 1,3-disubstituted allenes are amenable to the catalytic semihydrogenation, by preferentially hydrogenating the alkyl-substituted electron-rich C=C bonds in forming thermodynamically stable alkenes **3ar–3at** in good yields and high *E*-selectivity. The semihydrogenation of trisubstituted allenes proceeds effectively, selectively hydrogenating the less crowded and electron-rich C=C bonds in giving alkenes **3av**–**3ax**. The group of 1,3-dioxanyl can be tolerated by the

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Scheme 3. CAAC-phosphine-Cr-catalyzed semihydrogenation of di-, tri-, and tetrasubstituted allenes. [a] Conditions: **1** (0.4 mmol), **2g** (2 mol%), Mg (0.75 equiv), Me_3SiCl (0.5 equiv), THF (2 mL), H_2 (30 atm), 50 °C, 12 h.



Scheme 4. Scalability and catalytic applications.

hydrogenation system, opening up the opportunity in the synthesis of carboaldehyde derivative by late-stage functionalization (**3ay**). To examine the regio- and stereoselectivity, trisubstituted allenes containing two different substituents of aryl and methyl at the C_1 position of allenic scaffolds were synthesized and hydrogenated with CAAC-phosphine-Cr catalysis. We were pleased that *E*-alkenes **3az** and **3ba** could be selectively formed by hydrogenating less substituted C=C bonds, achieving good yields and high stereoselectivity. In addition to di- and trisubstituted allenic scaffolds, tetrasubstituted allenes are suitable in the semihydrogenation by the selective reduction of electron-donating alkyl-substituted C=C bonds in the formation of cyclopropyl-, cyclopentanyl-, cyclohexyl-, isopropyl-, and isobutyl-substituted alkene products **3bb**–**3bf**.

Catalytic Applications

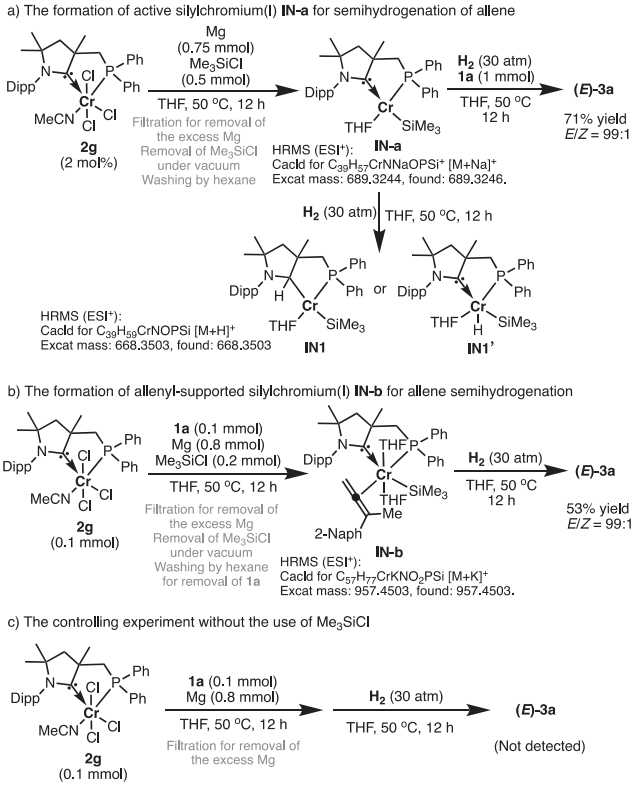
Inspired by these results, we next explored the scalability of the undirected semihydrogenation by CAAC-phosphine-chromium catalysis (Scheme 4a). It is evident that the semihydrogenation of allene could be conducted on gram scale without optimizing the catalytic conditions, giving *E*-alkene product (**3a**) in good yield and excellent regio- and chemoselectivity without over-hydrogenation. We expected that the semihydrogenation could be used for late-stage diversification of general semi-reduced *E*-alkenes. The cyclization of **3c** with diphenylmethanol or 2-aminobenzenethiol led to the formation of substituted benzocycles of 2,3-dihydro-1*H*-indene (**5**) and 3,4-dihydro-2*H*-benzo[*b*][1,4]thiazine (**6**) in nearly single diastereomers (d.r. > 19:1) (Scheme 4b). Oxirane **7**, which is substituted by dimethyl and phenyl scaffolds, was facilely formed by the oxidation of **3c**. The incorporated hydride moiety of *E*-alkene can be halogenated in forming alkenyl bromide **8**, which perfectly couples with aniline and chlorodiphenylphosphane to give alkenylated electroluminescent material **9** and phosphine ligand **10** (Scheme 4c). The late-stage functionalization by cross-coupling reactions with pharmaceuticals of acetaminophen and metaxalone formed the related derivatives **11** and **12**, offering the possibility for the development of new drugs and candidates.

Studying Silylchromium Intermediates and Mechanistic Insights

Since Wilkinson reported the first example of transition-metal–silyl complex ($Cp(CO)_2FeSiMe_3$) in 1953,[79] silyl-ligated metal complexes and intermediates have

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Scheme 5. Mechanistic studies in exploring silylchromium species.

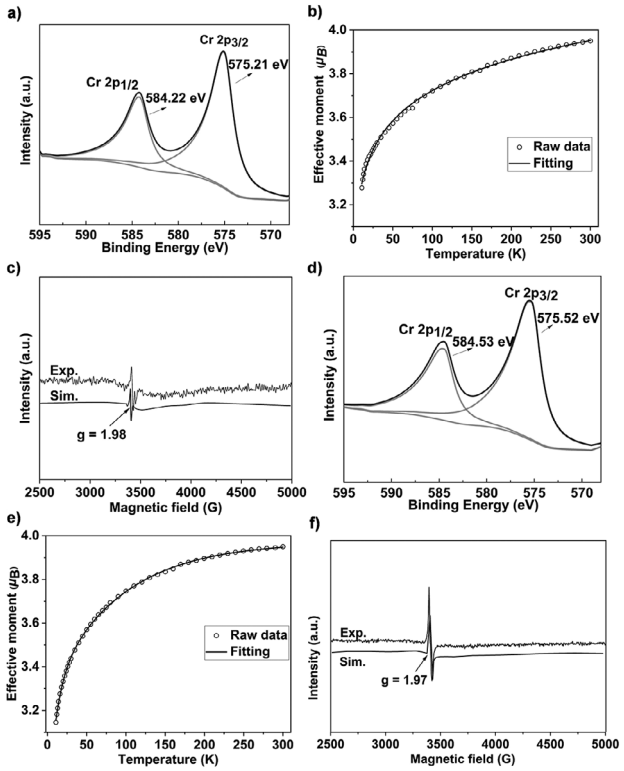


Figure 1. XPS and EPR spectra, and magnetic susceptibility changes for the silylchromium and related intermediates. a) XPS spectrum of **IN-a**. b) Temperature-dependent magnetic susceptibility plot for **IN-a**. c) EPR spectrum of **IN-a**. d) XPS spectrum of **IN-b**. e) Temperature-dependent magnetic susceptibility plot for **IN-b**. f) EPR spectrum of **IN-b**.

received great interest for their potential applications in catalytic transformations.[80–82] Recent studies suggested that transition-metal–silyl species could serve as valuable intermediates in the catalytic dehydrosilation,[83,84] olefin hydrosilylation,[85–87] C─H borylation and silylation,[88,89] and olefin hydrogenation.[90,91] Taking into consideration the requirement of chlorosilane for the undirected semihydrogenation of allenes by CAAC-phosphine-chromium catalysis, it comes as no surprise that the in situ formed low-valent CAAC-phosphine-chromium complex by reducing **2g** (2 mol%) with magnesium essentially reacts with Me_3SiCl, leading to the formation of silyl-ligated chromium species (Scheme 5a).[92,93] High-resolution mass spectrometry (HRMS) examination of the resulting complex suggested one prominent ion at a mass-to-charge ratio of 689.3246 (calculated m/z 689.3244) corresponded to that expected for CAAC-phosphine-supported (silyl)chromium(I) **IN-a** (see Supporting Information).[94] The analysis of Cr 2*p* binding energies by core-level X-ray photoelectron spectroscopy (XPS) shows two sets of peaks with the $2p_{1/2}$ component appearing at 584.22 eV and the $2p_{3/2}$ component appearing at 575.21 eV (Figure 1a). The Cr $2p_{3/2}$ binding energy of **IN-a** is virtually identical with that of the reported chromium(I) complex of $[Cr(CNPh)_6]PF_6$ (575.3 eV), indicating a monovalent chromium complex of **IN-a**.[95] The magnetic susceptibility of **IN-a** was tested at the temperatures from 300 to 10 K under 1000 Oe dc field. The effective magnetic moment was measured to be 3.95 μ_B at $T = 300$ K with $S = 3/2$ (Figure 1b). Combining with the result of XPS analysis, an intermediate-spin quartet state with the d^5 electronic configuration for this silylchromate(I) **IN-a** can be considered.[96] Upon cooling, the effective magnetic moment decreases slightly down to a minimum value of 3.28 μ_B at ca. $T = 10$ K, indicating some magnetic anisotropy. The electron paramagnetic resonance (EPR) measurements of **IN-a** were performed at X-band (9.8 GHz), showing a weak and sharp resonance at $g = 1.98$ ($\Delta H = 20$ G) (Figure 1c). Although no additional hyperfine splitting from ligands was shown, this spectrum is typical for intermediate-spin Cr(I) species.[97] Notably, this CAAC-phosphine-supported silylchromium(I) shows high activity in the semihydrogenation of **1a** under H_2, providing access to **(*E*)-3a** in 71% yield and 99:1 E/Z-selectivity (Scheme 5a).

By the further treatment of **IN-a** with H_2, the formation of the hydride intermediate, such as **IN1** or **IN1'**, was confirmed by HRMS ($[M + H]^+$; calculated $m/z = 668.3503$; found 668.3503) analysis (see Supporting Information). Munz and coworkers previously observed an impressive hydride migration from the Pd (^{fun}CAAC-Pd─H) to the carbene carbon of ^{fun}CAAC.[98] The theoretical studies suggested that the hydride transfer to the ^{fun}CAAC ligand from Cr center is energetically favorable.[75] These indicate that the ^{fun}CAAC-hydrided silylchromium(II) **IN1** could be considered by H-transfer to carbene center. Upon conducting the reaction of **2g** with **1a**, Me_3SiCl and magnesium, the analysis of the resulting complex by HRMS ($[M + K]^+$; calculated $m/z = 957.4503$; found 957.4503) and XPS (Cr $2p_{1/2}$,

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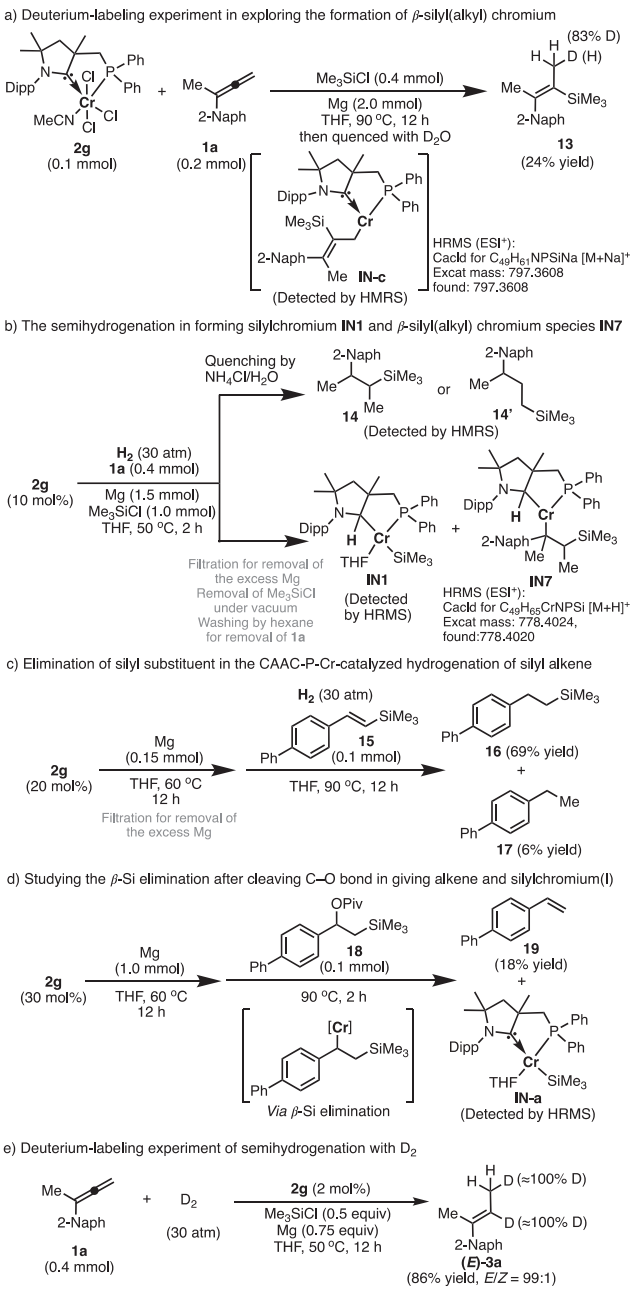
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584.53 eV; $2p_{3/2}$, 575.52 eV) indicates the formation of allene-coordinated silylchromium(I) intermediate of **IN-b** (Scheme 5b and Figure 1d).[95] The same effective magnetic moment at $T = 300$ K as that of **IN-a** was obtained by analysis of magnetism of **IN-b**, having the minimum value of 3.15 μ_B at ca. $T = 10$ K (Figure 1e). The EPR spectrum of **IN-b** displays a sharp resonance at $g = 1.97$ ($\Delta H = 25$ G) (Figure 1f). By hydrogenation with H_2, *E*-alkene product (**(*E*)-3a**) was formed in 53% yield and 99:1 *E*/*Z*-selectivity. In contrast, the controlling experiment by performing the reaction without chlorosilane suggests that the semihydrogenation does not occur (Scheme 5c). Collectively, the formation of CAAC-phosphine-ligated silylchromium as active species in the semihydrogenation of allenes could be considered.

Studying the Processes of Sila-Metalation of Allenes and β-Silyl Elimination and Mechanistic Insights

Thereafter, the addition of silylchromium to allene was investigated. The stoichiometric reaction of CAAC-phosphine-chromium complex **2g** with magnesium, allene and Me_3SiCl was conducted and quenched with D_2O. We found that the silyl-deuterated alkene compound **13**, by incorporating silyl group at the middle carbon of C=C=C bonds of allene, was formed in 24% yield with incorporation of 83% D (Scheme 6a). It suggests that the addition of silylchromium (**IN-a**) to the less crowded terminal C=C bond of allene occurs in a regioselective manner, leading to the formation of allylic intermediate such as **IN-c** by deuteration in forming **13**. We studied the related intermediates in the catalytic semihydrogenation of **1a**. By conducting the semihydrogenation with 10 mol% of **2g** for 2 h and quenching by NH_4Cl/H_2O, mass spectra recorded of solution suggest a hydrosilylated motif with an *m*/*z* 279.1540. The molecular formula $[C_{17}H_{24}SiNa]^+$ is consistent with the hydrosilylated alkane, such as **14** or **14′**, as depicted in Scheme 6b. Given the regioselective addition of silylchromium to allene by incorporating silyl scaffold at the middle carbon of C=C=C moiety in forming **13**, we proposed the formation of hydrosilylated alkane **14** in the reaction. It may be derived from a β-silyl(alkyl) chromium complex such as **IN7**. To test the possibility, the catalytic semihydrogenation was carried out for 2 h and the resulting residue was analyzed by HRMS. The mass spectrum suggests one prominent ion with an *m*/*z* = 778.4024, which may be designated to the β-silyl(alkyl) chromium species **IN7** that was formed by an initial sila-metalation of allene by addition of **IN1** to terminal C=C bond and hydride reduction in giving silylated alkene **13** and followed by hydrometallation with Cr–H species. In the reaction system, the hydride (silyl)chromate (e.g., **IN1**) can be detected by HMRS analysis.

As the mass spectra of analyzing the semihydrogenation of **1a** indicate the formation of β-silyl(alkyl) chromium species **IN7**, if this intermediate is involved in the semihydrogenation, the β-elimination of silyl group should occur to form the alkene product. Because there were no reports of β-silyl(alkyl) chromium complexes, we envisioned that the hydrogenation of silyl-substituted alkenes by CAAC-phosphine-Cr catalysis could provide evidence about the



Scheme 6. Deuterium-labeling experiments and studies of sila-metalation and β-silyl elimination processes.

β-silyl elimination. If the β-silyl elimination occurs after the addition of Cr–H species to alkene, the desilylated motif could be detected in the hydrogenation. By the hydrogenation of silylalkene **15** with CAAC-phosphine-chromium catalysis, we found that the desilylated alkane **17** can be formed and characterized by NMR spectra, albeit obtained in low yield (Scheme 6c). To further examine the process of β-silyl elimination, the reductive elimination of 2-(trimethylsilyl)ethyl pivalate **18** with CAAC-phosphine-Cr complex **2g** was performed (Scheme 6d). The related alkene compound **19** can be formed in 18% yield. Mass spectrometry data obtained from the reaction mixtures identify the silylchromium ion

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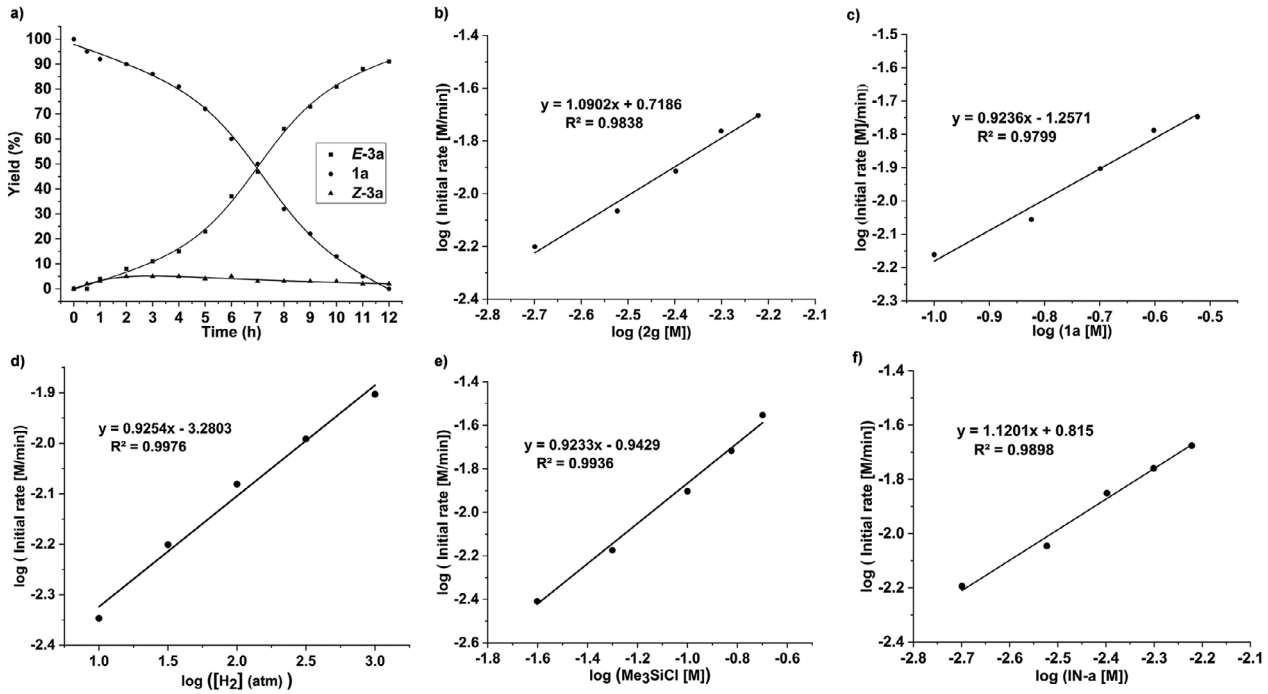


Figure 2. Kinetic studies. a) Reaction profile for selective semihydrogenation of allene of **1a** by using active CAAC-phosphine-chromium species that was formed in situ. b) Plot of the initial rate versus the concentration of CAAC-phosphine-chromium complex. c) Plot of the initial rate versus the concentration of allene **1a**. d) Plot of the initial rate versus the pressure of hydrogen. e) Plot of the initial rate versus the concentration of chlorotrimethylsilane. f) Plot of the initial rate versus the concentration of silylchromium species **IN-a**.

of **IN-a** with m/z = 689.3245, as expected for the cleavage of benzylic C─O bond and followed by β-silyl elimination in giving **19**. These findings suggest that the sila-metalation of allene forms the intermediacy such as **IN-c** and indicate that the subsequent hydride-reduction and hydrometallation might proceed, leading to the formation of β-silyl(alkyl) chromium intermediate **IN7** and conversion to *E*-alkene by process of β-silyl elimination. The deuterium-labeling experiment by performing the semihydrogenation under D_2 clearly suggested that there is no H/D exchange, indicating that process of β-hydride elimination might not be involved in the semihydrogenation (Scheme 6e).

Reaction Profile Studies

We sought to study the kinetics of the semihydrogenation of allene by CAAC-phosphine-chromium catalysis. Figure 2a shows profiles of the semihydrogenation carried out with 2 mol% of **2g**, suggesting that the semihydrogenation of **1a** to **3a** proceeds sluggishly within the initial 4 h to afford *E*-alkene in nearly 15% yield. Subsequently, the semihydrogenation accelerates its rate for forming *E*-alkene, giving ***E*-3a** in 89% yield with further 7 h. These suggest a mechanism that involves the slow formation of active chromium species in situ with an induction period during the semihydrogenation.

To understand the role of the reaction components in the turnover-limiting step, the kinetic studies of CAAC-phosphine-Cr-catalyzed semihydrogenation of allenes were performed. Standard semihydrogenation conditions used 1 equiv of allene (**1a**), hydrogen (30 atm), the active low-valent CAAC-phosphine-Cr that was formed in situ by reduction of precatalyst **2g** with magnesium, 0.5 equiv of Me_3SiCl, and THF as the solvent. The catalytic semihydrogenation was performed in a high-pressure autoclave, and the reaction progress was determined by GC analysis with an internal standard. Varying the concentration of CAAC-phosphine chromium complex **2g** (in the range of 1.0–3.0 mol%) showed a first-order dependence on the CAAC-phosphine chromium precatalyst, implying a single, monomeric chromium species involving in the turnover-limiting step (Figure 2b). Changing the initial concentration of allene (**1a**) and pressure of hydrogen led to a first-order dependence, indicating that the turnover-limiting step for the semihydrogenation is concentration-limited in allene and pressure-limited in hydrogen (Figure 2c,d). It was noted that the initial reaction rate (Δ[***E*-3a**]/Δt) against the initial concentration of chlorotrimethylsilane (in the range 0.025–0.2 M) showed a slope of 0.92 from the logarithm plot and linear fitting of the data, resulting in a first-order dependence and indicating that chlorotrimethylsilane might be involved in the turnover-limiting step (Figure 2e). Moreover, the initial reaction rate (Δ[***E*-3a**]/Δt) against the initial concentration of **IN-a** showed a slope of 1.12 from the logarithm plot and linear fitting of the data, resulting in a nearly first-order dependence (Figure 2f).

Theoretical Studies by DFT Calculations

To further understand the mechanism and selectivity of CAAC-phosphine-Cr-catalyzed semihydrogenation of

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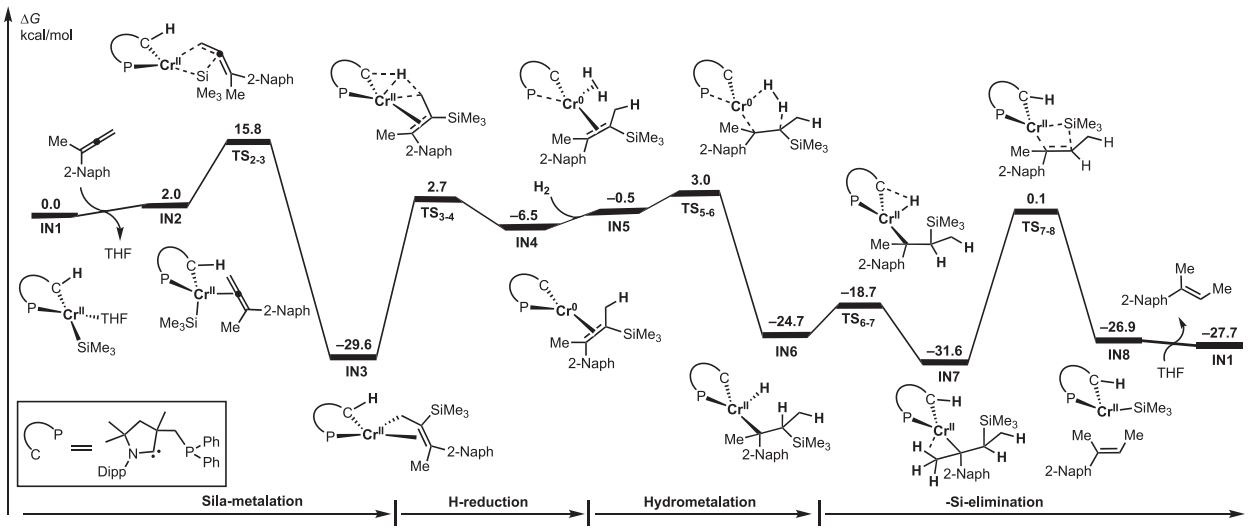


Figure 3. DFT studies carried out to determine the plausible sila-metalation and β-silyl elimination pathway involved in the selective semihydrogenation of allene **1a** by silylchromium catalysis. The quintet spin state, which keeps as the lowest one in energy for all the species, is shown in the figure.

allenes, density functional theory (DFT) modeling of the catalytic cycle was carried out using **1a** as substrate at the SMD-B3LYP-D3/def2-TZVP//B3LYP-D3/def2-SVP level of theory (Figure 3). Initiated from the experimentally detected ^{fun}CAAC-hydrided silylchromium **IN1**, its binding with **1a** by replacing the THF ligand would lead to allene-bound silylchromium **IN2**. By overcoming a relatively low barrier of 13.8 kcal mol^{-1} via $\mathbf{TS_{2-3}}$, the silyl group will migrate from chromium metal to the middle carbon of the C═C═C moiety. This sila-metalation process is highly exothermic, which is thus irreversible and generates the low-energy allylchromium intermediate **IN3** lying much below the reactant **IN2** by ~30 kcal mol^{-1}.[99] From **IN3**, the hydride on ^{fun}CAAC ligand can transfer directly to the terminal allyl carbon site to complete the H-reduction of the terminal C═C of allene via $\mathbf{TS_{3-4}}$, which needs to overcome a high barrier of 32.3 kcal mol^{-1}. Then the generated silylalkene in intermediate **IN4** would be hydrometalated in a very exothermic manner by H_2 via $\mathbf{TS_{5-6}}$, which would produce the alkylchromium-hydride species **IN6**. The subsequent hydride transfer from metal center to CAAC-phosphine ligand will take place to generate a more stable ^{fun}CAAC-hydrided β-silyl(alkyl) chromium intermediate **IN7**. Finally, by overcoming a high barrier of 31.7 kcal mol^{-1} via $\mathbf{TS_{7-8}}$, a β-silyl elimination will furnish the semihydrogenation of allene by giving the *E*-alkene product, and regenerate the active species **IN1** for closing the whole catalytic cycle. Notably, the three reaction steps via $\mathbf{TS_{3-4}}$, $\mathbf{TS_{5-6}}$, and $\mathbf{TS_{7-8}}$ have very similar barriers (32.3, 32.6, and 31.7 kcal mol^{-1}) with differences of less than 0.9 kcal mol^{-1}, which implies that all these transition states might influence the reaction kinetics. This pathway shown in Figure 3 is the only way that we found to be capable of yielding the *E*-alkene product directly, and no alternative pathway was found to generate *E*-alkene but *Z*-alkene instead (Figure S32), which is inconsistent with our experiments in stereoselectivity.

In Figure 4, to clarify the crucial β-silyl elimination process, we analyzed the Si─Cr bond-forming orbital for the

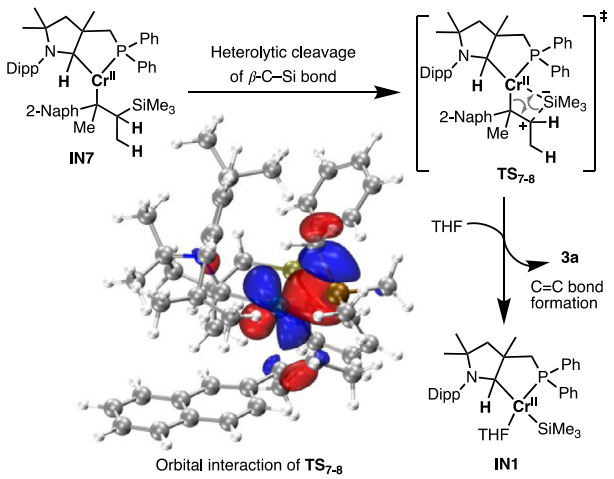


Figure 4. Orbital interaction diagram for the transition state ($\mathbf{TS_{7-8}}$) of β-silyl elimination in forming *E*-alkene.

corresponding β-silyl elimination transition state $\mathbf{TS_{7-8}}$. It can be seen that the doubly occupied *p* orbital of the silyl anion interacts strongly with the only one available empty 3*d* orbital of high-spin Cr(II) metal with d^4 shell, which is forming the dative Cr─Si bond. This picture of orbital interaction indicates that during the β-silyl elimination process, the C─Si bond is cleaved heterolytically but not homolytically. This heterolytic C─Si cleavage would generate carbon cation, which matches the anionic alkyl carbon on chromium metal in electron counting to generate the C═C π bond.

Conclusion

General strategy for enabling the semihydrogenation of undirected allenes to proceed in a chemo-, regio-, and stereoselective manner was developed based on cost-effective

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chromium catalysis. Using electron-rich and bulky CAAC-phosphine chromium(III) complex as precatalyst, the less crowded and electron-rich cumulated C=C bonds of di-, tri-, and tetrasubstituted 1,2-diene structures of allenes were selectively hydrogenated when the reactivity of CAAC-phosphine-ligated Cr(III) complex was enhanced by reduction with magnesium and reaction with chlorotrimethylsilane. Experimental and computational studies established the in situ formation of silylchromium as catalytically active species for the selective semihydrogenation of simple allenes. A wide range of unsaturated nitro, nitrile, alkenyl, ester, pyridinyl, furanyl, thiophenyl, pyrazolyl and hydrogenation-sensitive bromo, chloro, and fluoro substituents are tolerated by the hydrogenation system. Stoichiometric and catalytic reactions in exploring silylchromium intermediates, deuterium labeling and kinetic experiments, and computational studies support an initial sila-metalation of allene and a β-silyl elimination featuring heterolytic cleavage of C–Si bond to generate carbon cation, thus marching the anionic alkyl carbon on chromium metal in forming substituted C=C bond. It serves as a foundational example of experimental and theoretical elucidation of the effect of silyl as actor ligand on the selective hydrogenation catalyzed by high-spin transition metals.[100] Overall, the developed approach from this study provides a general and undirected avenue to chemo-, regio-, and stereocontrolled preparation of trisubstituted alkenes by semihydrogenation of allene, and improves the understanding of the factors for the development of synergistic silyl ligands for hydrogenation by abundant metal catalysis.[101]

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: β-silyl elimination · Allenes · Chromium catalysis · Semihydrogenation · Stereoselectivity

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